NSDDD 4.0 Search Interface Workshop



Searching 835 Documents from 125 Countries with Multilingual Semantic Embeddings

Andrew W. Neal | University of Edinburgh

Edinburgh National Security and Defence Documents Dataset v4.0 (pre-release, 2026)

What Is the NSDDD?

835
Documents

125
Countries

1987–2026
Time span

1,394,688
Text segments

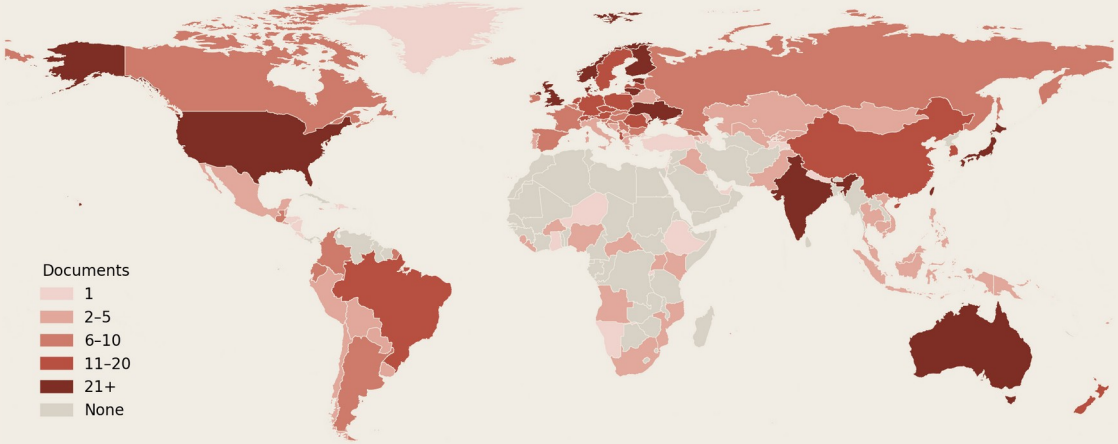
Document types: 354 Defence Documents | 223 National Security Strategies | 177 Defence White Papers | 81 Threat Assessments
Metadata: 27 international organisations, UN regions, Freedom House, World Bank income, ODA status, Huntington civilisations, small states classifications

https://andrewneal78.github.io/NSDDD_v4.0_installer/start_here.html or https://edin.ac/4shbP6M

NSDDD v4.0 is a pre-release. Until 4.0 is published, cite v3.5: Neal, A. W. and Gardner, R. B. (2026). NSDDD v3.5. University of Edinburgh. doi.org/10.7488/ds/8101

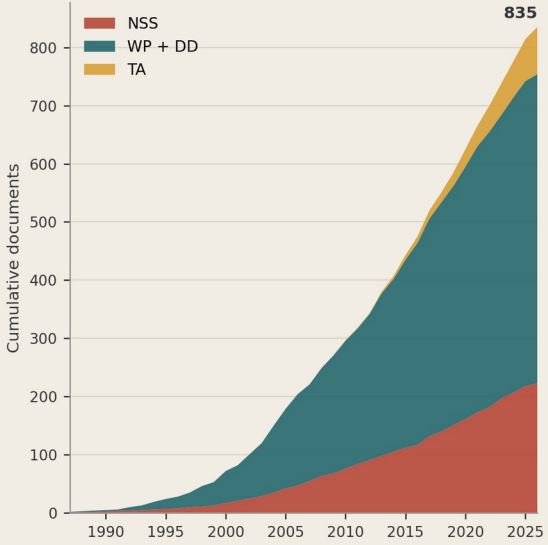
Coverage and Growth

Documents per country



125 countries and 835 documents, 1987–2026. Grey countries have no documents in the corpus.

Corpus growth



Cumulative, by publication year

What's In, What's Not

The published tip of the iceberg



In: top-level documents

National security strategies

Defence white papers and other top-level defence documents (reviews, directives, strategic plans)

Threat assessments (81), which describe threats more than they set policy

Not: sectoral and unpublished

Standalone cyber, maritime or counter-terrorism strategies

Anything a government has not made public, or that we could not locate

Horizontally comprehensive, not vertically specialised. See published document inclusion criteria.

What's New in NSDDD v4.0

A multilingual model and original-language search

Qwen3-Embedding-0.6B places 40 languages in one meaning-space: a query in any language finds passages in English and in the original languages (310,164 original-language segments in 39 languages). Each passage is counted once.

Corpus-wide agent re-cleaning

A whole-document generator+critic AI agent re-cleans every document: cleaner headings, finer segmentation, fidelity-gated against invention or loss. Every document also gains a clean Markdown rendering.

New documents, new countries

186 documents added and 22 removed (mostly duplicates found in an audit): 671 become 835. Seven new countries (Cabo Verde, Kosovo, Mozambique, Niue, Tajikistan, United Arab Emirates, Uzbekistan) take coverage to 125.

A new search interface and an MCP server

Numbered clusters with optional AI headings, keyword and Boolean search, Excel export with a settings sheet, update checks, and an MCP server so a command-line AI can run searches for you.

NSDDD v4.0 is a pre-release. The full release, with all the new 2026 documents and further additions, follows early in the new year. Figures may change.

What an Embedding Model Does

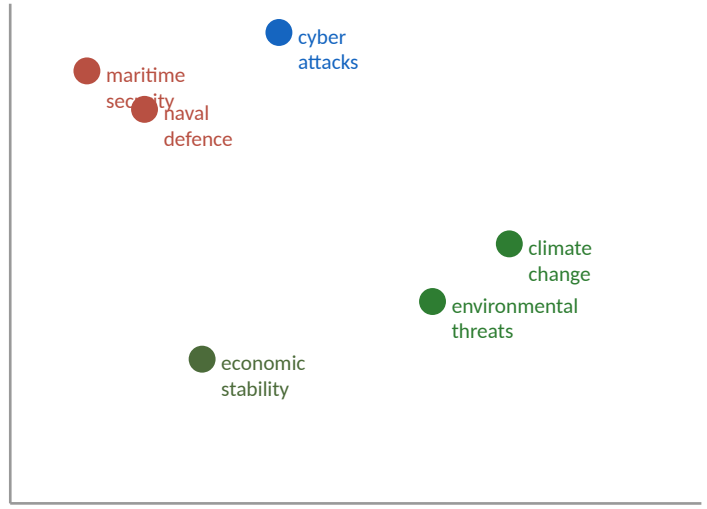
A GPS coordinate for meaning

Every sentence in the corpus, in English and in 39 other languages, has been passed through Qwen3-Embedding-0.6B, which turns it into a list of 256 numbers: a vector.

Just as latitude and longitude locate a place on earth, the vector locates the sentence in meaning-space. ‘Maritime security’ and ‘naval defence’ land close together; ‘climate change’ lands elsewhere.

The plot is a 2D simplification of 256 dimensions. Because the model is multilingual, a sentence and its translation land close together.

Simplified 2D meaning-space



Not generative AI: a sentence goes in, a vector comes out, never text. Results are always real sentences.

Semantic Search and Why Full Sentences Matter

The search pipeline

1. You type a query as a full sentence
2. The query is encoded into a 256-number vector
3. Compared against all 1,394,688 pre-encoded segments
4. Cosine similarity gives each segment a score
5. Results above the threshold (default 0.70) are ranked by score
6. Results are grouped into numbered clusters

Why full sentences?

The model was trained on sentence pairs. Single words produce imprecise vectors.

'cyber' → vague; matches everything digital

'cyber attacks on critical national infrastructure' → precise; finds hacking, infrastructure vulnerability, digital resilience

The sentence you type is a description of the meaning you are looking for.

For a specific word or phrase, use keyword search instead.

Think carefully about what meaning you are searching for. Different phrasings find different things.

Writing a Good Query

Three rules

1. Write a full declarative sentence describing the meaning you want, phrased as the documents would state it.

2. Do not ask a question. The system does not answer: it finds text close to your wording, and questions are rare in these documents.

3. One concept per query. A compound query averages two meanings and points at neither. Split it into separate searches and compare.

What to type instead

✗ “What do states say about climate change?”
✓ “Climate change poses a threat to national security”

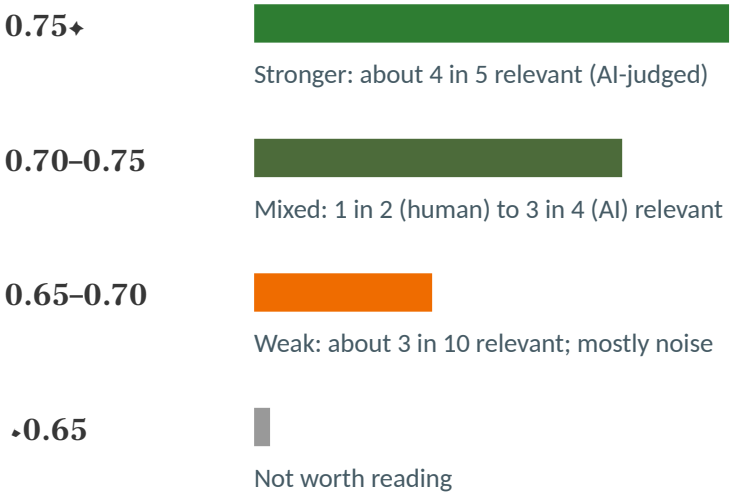
✗ “Climate change is a threat and cyber attacks are a growing risk”
✓ “Climate change is a threat to national security”
✓ “Cyber attacks on critical national infrastructure”

A query describes the meaning you are searching for. It is not a question to be answered.

One full declarative sentence, one concept, one query.

The Similarity Score

What the score means (provisional)



Choosing a cutoff

0.75 = evidence; 0.70 = exploring. The default, 0.70, is the bottom of the mixed band.

Relevant results can sit deep in the list: a hit at rank 80 (0.71) may beat rank 5 (0.83) for your question.

Score-based: review all above a threshold. Proportion-based: stop when most results are off topic. Manual: export and mark accept or reject.

Document your cutoff so the analysis is reproducible.

The bands vary by topic and filters. In one climate test only about half of the results at 0.75+ contained the stem 'climat' (a crude check on one query), so read results before relying on a band.

Three Search Modes

Semantic search (default)

Finds meaning, not words, in English and in 39 other languages. Best with one full declarative sentence. Compares your query against 1,394,688 pre-encoded segments.

Keyword search

Finds an exact word or phrase, including in original languages (Gesamtverteidigung). Wildcards: cyber* finds cyberspace and cybersecurity. Whole words only without a wildcard; optional fuzzy matching.

Boolean search

Tick keyword search, then ‘Enable Boolean operators’. AND, OR, NOT and brackets combine terms within one passage, e.g. (nuclear OR radiological) AND weapon, or terrorism AND NOT domestic. NOT needs AND before it.

All three modes group results into clusters. Keyword and Boolean results are exact matches, not scored by meaning.

Searching in Many Languages

Cross-lingual search

A query in any language matches passages in every language, because the model places a sentence and its translation close together. The corpus holds 1,084,524 English segments and 310,164 segments in 39 other languages (Norwegian, Spanish, German, Chinese, Portuguese, French, Korean …).

For 252 documents the original-language text is aligned to the English. ‘Search languages’ lets you search everything (default), English only, or original languages only.

Each passage counts once

When a passage matches in both its original language and its English translation, the two are merged into one English result (‘Count each passage once’, on by default). The number merged is shown above the results.

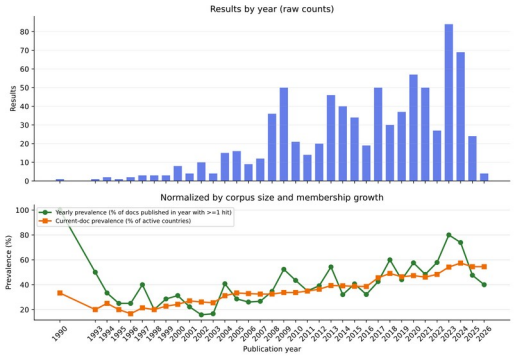
A few original-language results can remain: passages whose English did not match. They are genuine hits. Untick the box to count both sides.

Result links open the original and the English side by side.

Search in the language you think in. Counts are per passage, not per language version.

The Visualisation Outputs

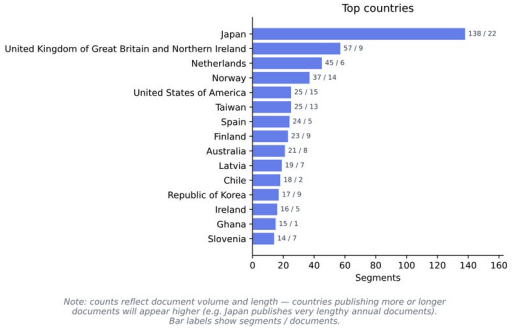
Year chart



Top: raw segment counts per year, which can mislead without context.

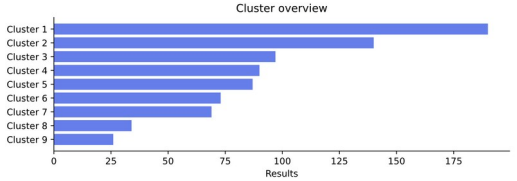
Green: share of that year's documents with a hit. Orange: share of active countries whose latest document mentions the topic.

Country chart



Top 15 countries by raw count, with an explicit caveat about document volume and length.

Cluster chart



How results group by meaning, numbered by size. The table gives each cluster's most typical sentence (hover to read it). AI headings are optional.

Clusters and AI Headings: Use with Care

What they do

Clustering groups the returned segments by meaning (Ward linkage on the embeddings), so you can see sub-themes without reading every segment. Clusters are numbered by size; the table gives each one's most typical sentence, and the 'Cluster detail' slider sets how fine the grouping is.

'Add AI headings' asks a small AI model on your own computer (Qwen3-0.6B) to write a 2 to 8 word heading for each cluster. A cluster it cannot name keeps its number.

Cautions

Neither changes the similarity scores or the export. Both are navigation aids, not analytical methods.

Cluster membership is a heuristic grouping, not a classification. Headings are generated text and can be wrong, vague or too general.

Check each heading against the cluster's example sentence (hover, or the 'Cluster example' column in the export) before you quote, count or code with it.

Clusters and headings help you navigate. Neither is an analytical conclusion.

Qualitative Work with the Exported Data

The Excel and CSV export

By default the export is an Excel workbook (.xlsx) with three sheets: Results, Search settings and Normalisation context (CSV is optional). Result columns: cluster, score, segment text, context before and after, document, year, country, ISO code, region, subregion, income group, freedom status, link, and the matched language with its original-language text. Add your own 'Accept' and 'Notes' columns for coding.

The whole-of-government example

A search for 'whole-of-government approach to security' found the full genealogy of this idea (traced in the NSDDD 3.5 export):

'Broad, holistic approach' (South Africa 1998) → 'comprehensive approach' (NZ 2000) → 'integrated/coordinated' (Canada, Singapore, Uganda, Finland 2004) → 'whole-of-government' crystallises (Australia 2005) → 'whole-of-society', 'whole-of-nation' (recent)

In 4.0, at a threshold of 0.72, the search returns 138 passages from 80 documents in 44 countries (at 0.75: 26 passages, 17 documents, 14 countries). Semantic search finds the variants because they are close in meaning, though they use different words; a keyword search would miss most of them.

Manually colour-coding the results in the exported spreadsheet revealed the range of formulations over geopolitical space and time.

The Export Workbook in Excel

	A	B	C	D
1	Topic	whole of government approach to security		
2	Search type	Semantic		
3	Search threshold	0.7		
4	Cluster threshold	0.78		
5	Number of search results	2560		
6	Number of clusters	27		
7	Context window	1		
8	Active filters	Years: 1987–2025		
9				
10	Cluster ID	Cluster Name	Search score	Segment text
11	cluster_0	Policy Strategy Security policy	0.8456	Guaranteeing a Whole-of-Government Approach to Security
12	cluster_0	Policy Strategy Security policy	0.8229	These circumstances necessitate an effective whole-of-government approach to security
13	cluster_0	Policy Strategy Security policy	0.8181	Whole of Government Responses Current threats to security require a whole-of-government approach
14	cluster_0	Policy Strategy Security policy	0.8097	A whole-of-government approach to security – dealing with emergencies, disasters and new threats is a task we all share
15	cluster_0	Policy Strategy Security policy	0.8094	A whole-of-Government approach towards national security is needed to reduce duplication of effort and to increase inter-sectoral coordination
16	cluster_0	Policy Strategy Security policy	0.8071	What is the relationship between Defence and other government agencies to enhance a 'whole of Government' approach towards achieving secur
17	cluster_0	Policy Strategy Security policy	0.8052	It sets out a 'whole of government' approach, based on a concept of security that goes beyond military effects
18	cluster_0	Policy Strategy Security policy	0.8027	The watchword of this Security Strategy is a whole-of-government and whole-of-society approach
19	cluster_0	Policy Strategy Security policy	0.796	This is why a resolute approach to ensuring security must be conceived and carried out in a whole-of-government manner
20	cluster_0	Policy Strategy Security policy	0.7936	It is important to note that the policy provides a whole-of-government approach to address and strengthen national security
21	cluster_0	Policy Strategy Security policy	0.7933	The authors applied the whole-of-government approach to the text, which has improved our understanding of the breadth and complexity of secur
22	cluster_0	Policy Strategy Security policy	0.7926	As a necessary foundation for an effective defence policy, the importance of whole- of-government approaches to the security of the State in its int
23	cluster_0	Policy Strategy Security policy	0.7923	A whole-of-government approach is needed to reduce and eliminate threats to national security
24	cluster_0	Policy Strategy Security policy	0.7913	The Government will work to promote a comprehensive approach to security
25	cluster_0	Policy Strategy Security policy	0.7904	As a whole of government approach, this review provides a common perspective for all elements of government involved in the security of the State
26	cluster_0	Policy Strategy Security policy	0.7899	This whole-of-government framework includes a wide range of government players in understanding the threats, planning responses and impleme
27	cluster_0	Policy Strategy Security policy	0.7844	We need a whole-of-government approach to implementing this National Security Strategy
28	cluster_0	Policy Strategy Security policy	0.7843	The comprehensive interpretation of the concept of security makes it indispensable for its political, military, economic, financial, environmental ar
29	cluster_0	Policy Strategy Security policy	0.7838	A MODERNISED TOTAL DEFENCE CONCEPT Our overall defence effort relies heavily on a whole-of-government approach
30	cluster_0	Policy Strategy Security policy	0.7837	e. Internal security Internal security and civil protection require a whole-of-government approach

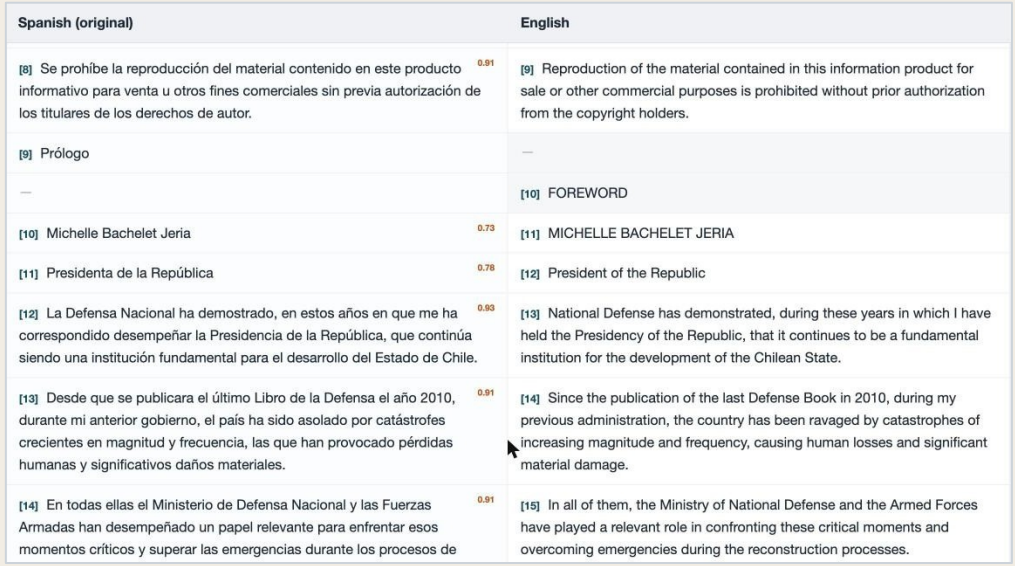
Screenshot from the NSDDD 3.5 export. The 4.0 export adds the matched language, the original-language text and a clickable link.

Direct Links to Source Segments

How the links work

Each result has a direct link that opens the segment in its full document context in the NSDDD viewer. Documents with an original-language version open as a parallel view, original and English side by side.

In Excel the links are clickable. In a CSV they are plain addresses. All links work only while NSDDD is running.



Spanish (original)		English
[8] Se prohíbe la reproducción del material contenido en este producto informativo para venta u otros fines comerciales sin previa autorización de los titulares de los derechos de autor.	0.91	[9] Reproduction of the material contained in this information product for sale or other commercial purposes is prohibited without prior authorization from the copyright holders.
[9] Prólogo		—
—		[10] FOREWORD
[10] Michelle Bachelet Jeria	0.73	[11] MICHELLE BACHELET JERIA
[11] Presidenta de la República	0.78	[12] President of the Republic
[12] La Defensa Nacional ha demostrado, en estos años en que me ha correspondido desempeñar la Presidencia de la República, que continúa siendo una institución fundamental para el desarrollo del Estado de Chile.	0.93	[13] National Defense has demonstrated, during these years in which I have held the Presidency of the Republic, that it continues to be a fundamental institution for the development of the Chilean State.
[13] Desde que se publicara el último Libro de la Defensa el año 2010, durante mi anterior gobierno, el país ha sido asolado por catástrofes crecientes en magnitud y frecuencia, las que han provocado pérdidas humanas y significativos daños materiales.	0.91	[14] Since the publication of the last Defense Book in 2010, during my previous administration, the country has been ravaged by catastrophes of increasing magnitude and frequency, causing human losses and significant material damage.
[14] En todas ellas el Ministerio de Defensa Nacional y las Fuerzas Armadas han desempeñado un papel relevante para enfrentar esos momentos críticos y superar las emergencias durante los procesos de	0.91	[15] In all of them, the Ministry of National Defense and the Armed Forces have played a relevant role in confronting these critical moments and overcoming emergencies during the reconstruction processes.

Document 69 (Chile, Defence Book 2017): Spanish original left, English right. Numbers are match scores; a dash means no confident match.

Each result links directly to the segment in its original document. Use these links. Context changes meaning.

Metadata Filtering: In-Tool and in Excel

In-tool filters (before searching)

- Region, subregion, individual countries
- 27 international organisations (membership as published, or today's)
- Income group, democracy, ODA status
- Small states, S. Huntington's 'civilizations'
- Document type, year range, search languages
- Most recent document per country only

Spreadsheet filters (after searching)

The export includes columns for country, ISO code, region, subregion, income group and freedom status.

You can sort and filter on any of these in a standard Excel table after the search.

For many use cases, this is simpler than pre-filtering.

When to use which

Use in-tool filters when you want to restrict the search scope (e.g. only NATO members when each document was published, or only post-2010 documents). This affects both results and performance.

Use spreadsheet filters when you want to explore and slice the full results after the fact.

Both are valid and complementary.

Read It Responsibly

1 Published, not complete

The dataset holds what states chose to publish and we could find. Absence is not absence of strategy.

2 Words, not deeds

Documents are political performances, not intelligence assessments or evidence of implementation.

3 Translated, uniformly

About 27% of documents are machine-translated. Translation suits bag-of-words methods (de Vries et al. 2018) but is untested for semantic search; check parallel text if concerned.

4 Genres blur

The four types are a guide. Boundaries are not watertight.

5 Retrieved ≠ relevant

A search result is a candidate. Put a human standard between it and any claim.

6 Mind the version

Documents and metadata change between versions. Cite v3.5 until 4.0 is published.

Acknowledge and work with the limits.

Cautions for Quantitative Work

Unequal document counts

Norway has 37 documents and Japan 29, while 32 countries have just one. Cross-country comparisons must account for this.

Unequal document length

Japan publishes exceptionally long annual defence documents (about 190,000 words on average). The system works at the sentence level, so longer documents contribute more sentences.

Unequal time coverage

Some countries have documents spanning decades; others entered the dataset recently (seven countries are new in 4.0). Time-series analysis needs care.

Sentence counts ≠ document prevalence

Raw counts can be pulled up by repeated mentions within a single document. Use document-level prevalence, not sentence counts, for trend analysis. Each passage is now counted once, so original-language and English matches no longer double count.

The document_metadata_4.0 CSV

835 documents

25 metadata columns

125 countries

What each row records

Identity file prefix (permanent number), document name

Place and time country, ISO and M49 codes, G77 membership, year

Type and size type (NSS, WP, DD, TA), length, segments, words

Source up to 11 language columns, ministry, source URL

Where to find it

Installer folder
document_metadata_4.0.csv

Windows package
data\metadata\document_metadata.csv

Edinburgh DataShare
with the full 4.0 release

Check the file before designing cross-national comparisons: coverage is uneven, and it shows exactly what you have.

Document Types

NSS

National Security Strategies

A country's overall statement of threats, goals and priorities. The most strategic of the four types.

223 documents

WP & DD

Defence White Papers and Defence Documents

Defence policy, strategy, reviews and doctrine. They overlap but differ: Defence Documents had the higher rate of threat language in 33 of the 44 countries with both (75%). Analyse them separately, or test before merging.

177 + 354 documents

TA

Threat Assessments

Included as top-level published documents. They describe threats more than they set policy: useful for threat analysis; check before including them in policy work.

81 documents

The types are a guide, not a watertight classification: boundaries blur, so check borderline documents.

Normalisation Context

Sort at will

You need to decide which metrics are meaningful for your analysis.

Topic: ‘Climate change is a threat to national security’ (4.0, threshold 0.72)

Japan has the most hits (140, in 24 documents) but also has the longest and most documents in these results.

France was the first to discuss this topic (1994) and has discussed it for longest (1994–2025).

The United Arab Emirates, Cook Islands, Ghana and the Maldives have the highest hits per 1,000 words, but each rests on one document: a possible measure of salience, to be handled with care.

Country	Hits	Docs matched	Docs in corpus	Hits per 1,000 words	Years
Japan	140	24	29	0.021	2005–2025
Netherlands	97	11	19	0.268	2007–2025
Norway	58	15	37	0.021	2004–2025
Australia	34	12	33	0.044	2007–2026
New Zealand	28	6	16	0.105	2010–2025
Taiwan	26	14	23	0.020	1996–2025
Finland	25	11	21	0.043	2001–2026
United Arab Emirates	2	1	1	0.533	2023
Cook Islands	6	1	1	0.455	2023
Ghana	15	1	1	0.335	2020
Maldives	5	1	1	0.334	2012

Normalisation context sheet from a 4.0 search. Top: seven countries with most hits. Shaded: highest hits per 1,000 words.

Letting an AI Analyse the Results

+ Can do

Read the exported results

Build analyses and charts with established packages (pandas, matplotlib, scipy), so every step can be inspected and rerun

Draft coding samples for human coders

Set up multi-model pipelines

i Must be given

The full metadata folder on Edinburgh DataShare, including data that changes over time (UK in the EU, who joined NATO when)

The inclusion criteria

Instructions to normalise by document count, length and time: specify the analysis, not just the question

That documents are what states publish, not what they do

That coverage is uneven, and no match is not evidence of absence

! Watch for

Assumptions slipped into the analysis: dropped outliers, rescaling, merged categories

Geopolitical assumptions: e.g. that G77 countries lack security policy capacity

Evaluate its analyses and interpretations critically

Use AI to write the code. Design the analysis yourself, and audit what it assumes.

Using a CLI AI to Analyse Results

What is a CLI AI?

A command-line AI assistant you run in a terminal alongside NSDDD. It can read your exported file, write and run analysis code and, through the MCP server, run NSDDD searches itself, with your permission. Claude Code, OpenAI Codex (ChatGPT), Gemini CLI and OpenCode all do this job; check current pricing and install steps.

Practical workflow

1. Export your results. 2. Open a terminal in that folder. 3. Start your CLI AI; give it the filename and document_metadata_4.0.csv. 4. Ask it to write Python to normalise counts, plot trends and run stats. Or let it search NSDDD itself through MCP (next slide).

Give it the context

It does not know the NSDDD corpus. Pass on the cautions from the previous slide: the DataShare metadata, the inclusion criteria, and what to normalise.

The AI writes the code. You make the methodological choices and supply the domain knowledge.

Searching NSDDD from a CLI AI (MCP)

What is MCP?

The Model Context Protocol (MCP) is a standard way to give an AI assistant tools. NSDDD includes a local MCP server: your CLI AI can run NSDDD searches itself and read the results directly.

The server runs the same search code, thresholds, filters and export format as the search page.

The search runs locally, but the passages the AI reads are sent to the AI provider.

What the AI can do

Semantic and keyword search, with filters, clustering, context and optional export

List the available regions, organisations, years and document types

Report corpus, model and language statistics

Follow built-in guidance on query writing, score bands and counting each passage once

Tell your CLI AI to read and follow the NSDDD guidelines built into the MCP server, before it searches.

MCP lets an AI run the search for you. You still decide the question, the threshold and what counts as evidence.

Letting an AI Run the Search (via MCP)

+ Benefits

Composes many wordings of the same search topic

Can be good robustness test

Tries them all in minutes, not hours

Handles the repetitive work: searching, filtering, counting, exporting

! Downsides

May choose settings you do not notice: threshold, mode, filters, number of results

Check what it actually ran

You still have to justify the topics and the wordings

Who does what

You: Pose the question and justify the topic → **The AI: Writes many wordings and runs the searches** → **You: check the settings and queries, then judge**

Let the AI widen the search. You stay accountable for the question, the settings and the wordings.

Try It Yourself

What to search

What concept are you investigating in your own research? Type it as a full sentence.

Topic areas that work well:

- Threat framings: terrorism, cyber, climate, pandemics, hybrid warfare, energy, food, migration
- Institutions: civil-military relations, whole-of-government, oversight, interagency
- Strategic concepts: deterrence, resilience, alliances, sovereignty
- Regional: Arctic, Indo-Pacific, Euro-Atlantic, African peace architecture

Immediate things to try

- Watch the similarity scores shift as you move down the results
- Add AI headings and check them against each cluster's example sentence
- Filter by region or organisation and compare
- Sort by different columns of the Normalisation context sheet
- Open the Excel file and colour-code results by hand to trace terminological variation
- Try the same concept as both a sentence query and a keyword
- Search in another language, or tick 'Original languages only'

The goal is not a finished analysis. It is to develop an intuition for how meaning-based search differs from keyword search.

Getting NSDDD 4.0 and Updates

The search tool

A separate program built on the dataset. Download the installer from GitHub (QR code or link below), unzip it, and run Setup.bat (Windows) or Install NSDDD.command (Mac).

Needs 8 GB of memory, 10 GB of disk space, and Windows 10 or 11 or macOS 12 or later. Updates install automatically when you start it online.

The dataset

NSDDD 4.0 will be archived on Edinburgh DataShare: the original PDFs, plain-text versions in English and the original language, segmented text, and the metadata files.

Full release: early 2027 (835 documents, 125 countries, 1987–2026).

Until then, cite v3.5: doi.org/10.7488/ds/8101

Edinburgh DataShare

Search tool: github.com/andrewneal78/NSDDD_v4.0_installer/releases

Summary

Key Takeaways

1. The system uses multilingual semantic embeddings to find meaning across 835 documents in 39 languages.
2. Full sentences work better than single words. Use 0.75 for evidence, 0.70 to explore, and run several differently worded queries.
3. Quantitative work requires normalisation: document counts, lengths, and time coverage are unequal.
4. Qualitative work benefits from additional work on exported files. Check AI cluster headings against their example sentences.
5. An AI can run searches (through MCP) and write analysis code reliably, yet without care, can ignore vital context.

NSDDD v4.0 is a pre-release. Until the 4.0 record is published, cite v3.5: Neal, A. W. and Gardner, R. B. (2026). NSDDD v3.5. University of Edinburgh. doi.org/10.7488/ds/8101